

ASSIGNMENT 4

①

$$A = \begin{bmatrix} 7 & 2 & -3 \\ 2 & 5 & -3 \\ 1 & -1 & -6 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - \frac{2}{7} R_1$$

$$R_3 \rightarrow R_3 - \frac{1}{7} R_1$$

$$A' = \begin{bmatrix} 7 & 2 & -3 \\ 0 & 21/7 & -15/7 \\ 0 & -9/7 & -39/7 \end{bmatrix}$$

$$\frac{A_{21}}{A_{11}} = \frac{2}{7}$$

$$R_2 \rightarrow R_1 - \frac{2}{7} R_1$$

$$\begin{array}{rrr} 2 & 5 & -3 \\ \frac{2}{7} & \frac{5}{7} & -\frac{3}{7} \\ - & - & + \\ \hline 0 & \frac{21}{7} & -\frac{15}{7} \end{array}$$

$$\rightarrow \frac{A_{31}}{A_{11}} = \frac{1}{7} \Rightarrow R_3 \rightarrow R_3 - \frac{1}{7} R_1$$

$$\begin{array}{rrr} 1 & -1 & -6 \\ \frac{1}{7} & -\frac{1}{7} & -\frac{6}{7} \\ - & - & + \\ \hline 0 & -\frac{9}{7} & -\frac{39}{7} \end{array}$$

$$\rightarrow \frac{-9/7}{21/7} = -\frac{9}{31} \Rightarrow R_3 \rightarrow R_3 - \left(-\frac{9}{31}\right) R_2$$

$$\begin{array}{rrr} 0 & -\frac{9}{7} & -\frac{39}{7} \\ 0 & -\frac{9}{7} & \frac{135}{21} \\ - & + & - \\ \hline 0 & 0 & -\frac{192}{31} \end{array}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{27} & 1 & 0 \\ \frac{1}{9} & -\frac{9}{31} & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 7 & 2 & -3 \\ 0 & \frac{31}{7} & -\frac{15}{7} \\ 0 & 0 & -\frac{192}{31} \end{bmatrix}$$

$$\textcircled{b} D = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{27} & 1 & 0 \\ \frac{1}{9} & -\frac{9}{31} & 0 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 18 \\ -6 \end{bmatrix}$$

$$\boxed{LD = B} \quad d_1 = 12$$

$$\frac{2}{7}(12) + d_2 = 18 \Rightarrow d_2 = 18 - \frac{24}{7} = \frac{102}{7}$$

$$\frac{1}{9}(12) - \frac{9}{31}\left(\frac{102}{7}\right) + d_3 = -6 \Rightarrow d_3 = -6 - \frac{12}{7} + \frac{418}{217} = -108/31$$

$$\rightarrow \boxed{UX = D} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad -\frac{192}{31}x_3 = -\frac{108}{31} \Rightarrow x_3 = \frac{108}{192} = \frac{9}{16}$$

$$\boxed{x_3 = 0.5625}$$

$$\rightarrow \frac{31}{7}x_2 - \frac{15}{7}\left(\frac{9}{16}\right) = \frac{102}{7}$$

$$\Rightarrow \frac{31}{7}x_2 = \frac{102}{7} + \frac{135}{112} \Rightarrow \boxed{x_2 = \frac{57}{16} = 3.5625}$$

$$\Rightarrow 7x_1 + 2\left(\frac{57}{16}\right) - 3\left(\frac{9}{16}\right) = 12 \Rightarrow 7x_1 + \frac{87}{8} = \frac{192}{8}$$

$$\Rightarrow x_1 = 15/16 = 0.9375.$$

② we will use $B \rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$$B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad D_1 = \begin{bmatrix} 1 \\ -\frac{2}{7} \\ -\frac{7}{31} \end{bmatrix}$$

$$U = \begin{bmatrix} 7 & 2 & -3 \\ 0 & \frac{31}{7} & -\frac{15}{7} \\ 0 & 0 & -\frac{192}{31} \end{bmatrix}$$

$$D_1 = \begin{bmatrix} 1 \\ -\frac{2}{7} \\ -\frac{7}{31} \end{bmatrix}$$

$$x_1 = \begin{bmatrix} x_{11} \\ x_{21} \\ x_{31} \end{bmatrix}$$

$$-\frac{192}{31} x_{31} = -\frac{7}{31}$$

$$x_{31} = \frac{7}{192}$$

$$\frac{31}{7} x_{21} - \frac{15}{7} x_{31} = -\frac{2}{7} \Rightarrow \frac{31}{7} x_{21} - \frac{15}{7} \left(\frac{7}{192} \right) = -\frac{2}{7}$$

$$\frac{31}{7} x_{21} - \frac{15}{7} \left(\frac{7}{192} \right) = -\frac{2}{7}$$

$$31x_{21} - 15 \left(\frac{7}{192} \right) = -2$$

$$31x_{21} - \frac{105}{192} = -\frac{384}{192}$$

$$31x_{21} = \frac{-384 + 105}{192} = \frac{-279}{192}$$

$$x_{21} = -\frac{9}{192}$$

$$7x_{11} + 2x_{21} - 3x_{31} = 1$$

$$7x_{11} + 2\left(-\frac{9}{192}\right) - 3\left(\frac{7}{192}\right) = 1$$

$$7x_{11} - \frac{18}{192} - \frac{21}{192} = 1 \quad \Rightarrow \quad 7x_{11} - \frac{39}{192} = \frac{192}{192}$$

$$x_{11} = \frac{33}{192}$$

$$x_1 = \begin{bmatrix} 33/192 \\ -9/192 \\ 7/192 \end{bmatrix}$$

$$A^T = \frac{1}{192} \begin{bmatrix} 33 & -15 & -9 \\ -9 & 39 & -9 \\ 7 & -9 & -31 \end{bmatrix}$$

$$\textcircled{2} \quad \max(|17| + |21| + |-3|, |21| + |5| + |-3|, |11| + |-1| + |-6|) =$$

$$\max(12, 10, 8) = 12$$

$$\max\left(\frac{57}{192}, \frac{57}{192}, \frac{47}{192}\right) = \frac{57}{192} = 0.296875$$

$$\begin{array}{r} 3.9375 \\ 12 \times 63 \overline{) 1560} \\ \underline{192} \\ 160 \\ \underline{192} \\ 160 \\ \underline{192} \\ 160 \\ \underline{192} \\ 160 \end{array} = \boxed{3.9375}$$

$$\textcircled{2} \quad A = L L^T \quad L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$$

$$A = \begin{bmatrix} 8 & 20 & 15 \\ 20 & 80 & 50 \\ 15 & 50 & 60 \end{bmatrix}$$

$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix} =$$

$$: \begin{bmatrix} l_{11}^2 & l_{11} l_{21} & l_{31} l_{11} \\ l_{21} l_{11} & l_{21}^2 + l_{22}^2 & l_{21} l_{31} + l_{22} l_{32} \\ l_{31} l_{11} & l_{31} l_{21} + l_{32} l_{22} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{bmatrix}$$

$$l_{11} = \sqrt{a_{11}} = \sqrt{8} = 2\sqrt{2}$$

$$l_{21} = \frac{a_{21}}{l_{11}} = \frac{20}{\sqrt{8}} = \frac{20}{2\sqrt{2}} = 5\sqrt{2}$$

$$l_{31} = \frac{a_{31}}{l_{11}} = \frac{15}{\sqrt{8}}$$

$$l_{22} = \sqrt{a_{22} - (l_{21})^2} \\ = \sqrt{80 - \left(\frac{20}{\sqrt{2}}\right)^2}$$

$$l_{22} = \sqrt{80 - 50} = 5\sqrt{2} = \sqrt{80 - \frac{400}{2}}$$

$$\underline{l_{32}}: l_{32} = \frac{a_{32} - (l_{31} \cdot l_{21})}{l_{22}}$$

$$l_{32} = \frac{50 - \left(\frac{15}{\sqrt{8}} \times \frac{20}{\sqrt{8}} \right)}{\sqrt{30}}$$

$$= \frac{50 - \left(\frac{300}{8} \right)}{\sqrt{30}}$$

$$l_{32} = \frac{50 - 37.5}{\sqrt{30}} = \frac{12.5}{\sqrt{30}}$$

$$l_{33} = \sqrt{a_{33} - l_{31}^2 - l_{32}^2}$$

$$l_{33} = \sqrt{60 - \left(\frac{15}{\sqrt{8}} \right)^2 - \left(\frac{12.5}{\sqrt{30}} \right)^2} = \sqrt{60 - \frac{225}{8} - \frac{625}{120}}$$

$$l_{33} = \sqrt{60 - \frac{675}{24} - \frac{125}{24}} = \sqrt{60 - \frac{800}{24}}$$

$$l_{33} = \sqrt{60 - \frac{100}{3}} = \sqrt{\frac{80}{3} - \frac{100}{3}} \Rightarrow l_{33} = \sqrt{\frac{80}{3}}$$

$$L = \begin{bmatrix} \sqrt{8} & 0 & 0 \\ \frac{20}{\sqrt{8}} & \sqrt{30} & 0 \\ \frac{15}{\sqrt{8}} & \frac{12.5}{\sqrt{30}} & \sqrt{\frac{80}{3}} \end{bmatrix}$$